

Introduction to EPIB 704

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EPIB 704

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Outline

1. Introductions
2. Course description
3. Setting the Baseline
4. The Generative AI Learning Penalty
5. Descriptive epidemiology
6. Reproducible research

Who are we?

Mabel

- Social and Infectious Diseases epidemiologist
- **"Recovered"** clinician; **ES(T)L**
- ***Twitter/FB/IG-less***
- Website

Sammy Lowe

Shirly Li

Some medical research bingo we will cover

unclear analysis aims causality, prediction or description: it makes a difference	evidence of absence fallacy not significant = no effect? if only things were that easy	data dredging “look what I found that turned out significant”	noisy data fallacy that what doesn’t kill it makes it stronger? i wish!
dichotomania categorize everything, chapter 1 of data torture handbook	table 2 fallacy interpret each regression coefficient as adjusted for confounding? yeah, that is probably wrong	regression to the mean yes, measurements at follow-up are different than at baseline. doesn’t mean much	ignoring dependent observations the data are probably nested, and that is important
poor reporting yes, there are reporting guidelines for that	small sample size gets these confidence intervals nice and wide	ignoring missing data can’t miss what you never had? think again	data driven variable selection post selection inference, yikes!
only apparent predictive performance that is optimistic. could have tried a cross-validation or a bootstrap, perhaps?	point estimate is the effect why did you even bother calculating the confidence interval?	collider is it a cause? is it a confounder? no, it is a collider here to ruin your inferences!	multivariate model what you mean to say is: <i>multivariable</i> model

Who are you?

What are the things you dislike the most about
EPI? 🤮

Course description

- Emphasis on epidemiologic theory and the estimation of epidemiologic effect measures and uncertainty in different study designs
- Emphasis on causal inference concepts and its distinction from, and yet simultaneously dependency, on statistical models
- Emphasizes the limitations of mainstream null hypothesis statistical significance (NHST) paradigm
- Encourages the recognition of the value of an estimation (Bayesian) paradigm for optimal inferences

Course description - more specifically

- Attention will also be given to exploratory data analysis (tabular and graphical), data interpretation, critical examination of assumptions and reproducible research
- Focus mainly on cumulative incidence measures for categorical outcomes, with attention to model checking and screening for confounders and effect measure modifiers
- Regression models covered will include linear, logistic, Poisson, survival, & meta-analytical (hierarchical) models
- Experimental and quasi-experimental designs will be discussed
- Miscellaneous topics will include attributable fractions, selection bias, sensitivity analyses, bootstrapping, matched data, missing data, and misclassified data

Objectives:

- 1) Identify common threats of the validity or biases,
- 2) Identify and interpret their effects in the measurement of associations, and
- 3) Identify opportunities to investigate, correct, and address these biases.

Learning Outcomes

By the end of this course you should be able to:

1. Contrast advantages and disadvantages of Frequentist and Bayesian statistical inference
2. Recommend analytical approaches to each (every) different study designs
3. Identify the presence of potential threats to the validity of a statistical analysis
4. Design a basic statistical analysis plan to answer a given research question accounting for threats to the validity

Instructional Methods

- “Hands-on” approach whereby the various key statistical concepts covered will be illustrated by computer coding, the new calculus for modern epidemiologic methods
- Counting has historically been the essential background to epidemiologic research and remains so in the 21st century but is now often best accomplished by simulation and sampling of posterior probability distributions
- Examples of the required computing code will be extensively provided.

Reference textbooks

- 1) ***What if?*** by Miguel Hernán and James Robins, available [here](#) (Chapters 1-11)
- 2) ***Modern Epidemiology*** by Timothy L. Lash, Tyler J. VanderWeele, Sebastien Haneuse, Kenneth J. Rothman, available at the McGill Library [here](#)
- 3) ***Regression and Other Stories*** by Andrew Gelman, Jennifer Hill, and Aki Vehtari, available [here](#)

Optional deeper Bayesian dive consider the excellent

- 4) ***Statistical rethinking: a Bayesian course with examples in R and Stan*** by Richard McElreath available at the McGill Library [here](#) (Chapters 1-10)

- Associated website: <https://xcelab.net/rm/statistical-rethinking/> and YouTube lectures [here](#)

Other readings will be assigned each week from published journal articles

Computing language

The computing language of choice for the course is [R](#).

There are several reasons for this choice including its open source and rich online community which means help is often only a [Google](#) away.

We believe [R](#) has become the *lingua franca* for much of the epidemiology/biostatistical universe. Of course, other languages including [Julia](#) or [Python](#) will give the same results but unfortunately, we can't supply the necessary support for those languages.

[Stata](#) is a popular proprietary software, but we consider the scripting and reproducibility offered by [R-Markdown](#) provides additional advantages for its choice.

R - Related resources

We understand that not everyone is familiar with [R](#) as a computing language and therefore **strongly recommend** starting, reviewing and or strengthening your knowledge on the software.

In-person: McGill's [Computational and Data Systems Initiative R Summer Camp](#): August 24 to 28, 2026.

Other resources from the [CDSI here](#)

R - Related resources

Online

- R-Studio Education: Independent learning (<https://education.rstudio.com/learn/>)
- R-Studio/posit: Practice-based learning (<https://posit.co/products/enterprise/academy/>)
- R-related Books: <https://www.rstudio.com/resources/books/>
- Princeton site to explore R: <https://exploringr.princeton.edu/self-learning-resources-for-r/>
- R-resources list: <https://thatdatatho.com/r-resources-beginner-advanced/>

Course Structure

- Lectures (Typical lectures, Seminars and Discussions!)
- Assignments (submitted through *mycourses*)
- Rapid Reviews

Assessment/Evaluation

	%	Rubrics
Attendance & Participation	5%	Presence, attention, active participation 📱😴 👓
Homework Assignments (4 x 12.5% + 1 x 20%)	70 %	Accuracy, demonstrated skills, timely delivery 📅⌚📈👤
Rapid Review (RR)	25%	Critical appraisal, time management ⌚📝🔬 🔍🦜
Total	100 %	

Attendance & Participation

We expect to see everyone **in class in-person** .

- Since life happens: *Zoom* attendance can be an option **under special circumstances ONLY**.

Attendance will be graded of 5 points to make up for the 5% of the total grade:

- Attend class, pay attention to the lectures and your classmates' presentations, and to participate actively of the in-class discussion.

Doing the suggested lecture **readings BEFORE the lectures** will assure an optimal learning experience.

Attendance & Participation

*To safeguard the well being of everyone, **if you present respiratory-like symptoms or any signs or symptoms of communicable conditions** that may endanger your health, the health of classmates or instructors, **please consult a medical professional and inform the instructors as early as possible** to make the required accommodations.*

Please see McGill's guidelines for attendance: [Guidelines and Policies](#)

Rapid Review

Every enrolled student must sign up for one *rapid review* which is a critical appraisal exercise consisting on:

- In-class Quiz and report (10/25)
- A 5 minute oral intervention (5/25)
- A short report (5/25)
- Interaction with the reading (5/25)

All based on a **methodological critique** about any new paper (**publication date since July 2026**) published in the biomedical literature (i.e., indexed in PUBMED).

The comments can be praises or critiques but **must** demonstrate thoughtful application of the principles discussed in the lectures as applied to new work in the field.

Graded out of 25 points to make up for the 25% of the total grade.

Guidelines for the *rapid review* (1)

- **Selection of the document:** Identify the published manuscript considering the date of publication, your personal/thesis interest and the methodological approach. **[0/25 total points]**

Consider the **focus on methods**, consider exploring new methodological approaches or novel use of existent epidemiological methods.

- E-pubs or online publications ahead and *pre-prints* are allowed.
- Please provide a pdf of your target paper **at least two weeks before the presentation** to be shared with the class.

Check with the instructor the suitability of the paper. Also, if having difficulties choosing, the instructor can help and could assign a paper instead

Guidelines for the *rapid review* (2)

- **All students** must participate in the review of the paper through the [Perusall platform](#).
 - You must have received an email to access the EPIB 704 Perusall platform.
 - You could also access it through this [link](#)
- This activity will have an individual grade of 5 points to make **5% of the total grade**.

Guidelines for the *rapid review* (3)

- **Critical appraisal:** Read and critically appraise the selected manuscript considering key methodological aspects of the design and analytic plan. Graded as **15/25 points** .

To illustrate understanding of the material, **EVERYONE must:**

- Answer a **structured quiz assessing the methodological aspect of the paper reviewed** and inquiring on specific epidemiological concepts discussed in the paper.**[10/15 total points]** **The resulting answer-sheet will serve a repository for consultation**
- Then, provide a **brief statement (150-words) indicating your thoughts and considerations about the methodological approach used** in the reviewed paper.
- Submit the **written document (max 150 words total)** the day of your review through myCourses's rapid review section. **[5/15 total points]**

Guidelines for the *rapid review* (4)

In-Class presentation format:

In a maximum total time of **5 minutes** present to the class the results of your review **[5/25 total points]**, considering the following:

- Power point / slides or any visual aids are welcome, but there **should not be more than 5 slides/pages**.
- After each presentation there would be a ***brief period for Q&As (approximately 5 minutes)***.
- Interactive discussions are highly encouraged but a **maximum of 15 minutes** are allocated for each presentation and discussion in every class.

Dates assignment for thr Rapid Review

select an article and a date (last date Nov 24) ¹ and **inform the instructor by September 8, 2026** 1st come 1st served.

Date	Name	Article/Manuscript
15 Sep	TAs	TBD
17 Sep		
...		
...		
...		
24 Nov		

Homework Assignments

The homework will make **70% of the total grade**. There will be 5 homework assignments (4 worth 12.5% + a final HW worth 20%) throughout the 13 weeks of the semester, which will be graded by the TAs. The schedule of dates for when these will be assigned and handed in and handed back is as follows:

HW #	Assigned	Due	Returned
1	Sept 15	Sept 25	Oct 1
2	Sept 29	Oct 08	Oct 15
3	Oct 15	Oct 27	Nov 3
4	Oct 29	Nov 10	Nov 17
5	Nov 10 *	Nov 24	Dec 1

* Last assignment is given 15 working days in advance. See Policy on Assessment of Student Learning (PASL) subsection 6.7. in Point 6. [COMMUNICATION OF COURSE ASSESSMENT TASKS](#).

How good are we at getting things done?



Howework Grading

Submitted assignments should be your **individual effort**, even if you consult with other students about your strategy for obtaining these solutions (see plagiarism note below).

To reinforce the concept of scientific **reproducibility** all assignments should be submitted as **R-Markdown** files. Data files for the assignments are available on GitHub and may be installed directly into **R**.

Late assignments carry a very severe penalty of **20% off per day late**. This is primarily to protect the time of the TAs.

Pleas of mercy for extenuating circumstances will be accepted only with written documentation. Taking extra time to do a better job is probably not a worthwhile strategy, because the late penalty is so costly.

Some Admin & Logistics

- Important dates (**McGill**): Add/Drop class: Sep 15th, 2026
 - Key dates: Assignments and Rapid Reviews
- Communication channels: *mycourses* & emails : Be Mindful of time schedules, office hours
 - We all have other activities outside the classroom
- **Course Evaluations** : Feedback throughout the course is MORE THAN WELCOME!

Read the syllabus! :)

What if I don't make it to the 65%?

1. **Supplemental assessment:** The student may complete supplemental work. The supplemental work will account for **20% of the course grade** (i.e., replacing the two final homework assignments) and will normally be **scheduled in early February 2027**
2. **Re-taking the course:** Alternatively, the student may choose to re-take the course in the following academic year.

BUT WE ARE MOSTLY AVOIDING THIS! By being proactive and finding strategies to improve early on!

Some Admin & Logistics

- [Academic Integrity](#) : Plagiarism, misconducts, & any form of discrimination, aggressive behavior or harassment towards ANY member of the community is **UNACCEPTABLE**
- [Policy for the academic accommodation of religious observances](#)
- [Guidelines for the Academic Accommodation of Pregnant Students and Students Caring for Dependents](#)
- [Policy concerning the rights to academic accommodations for students with disabilities](#)
- [Charter of Students' Rights](#)
- Other resources at [Student Wellness Hub](#) : FEEL FREE TO ASK!

Baseline Assessment Quizz



Where are we on our Epi knowledge?

The Generative AI Learning Penalty

● July 08, 2026

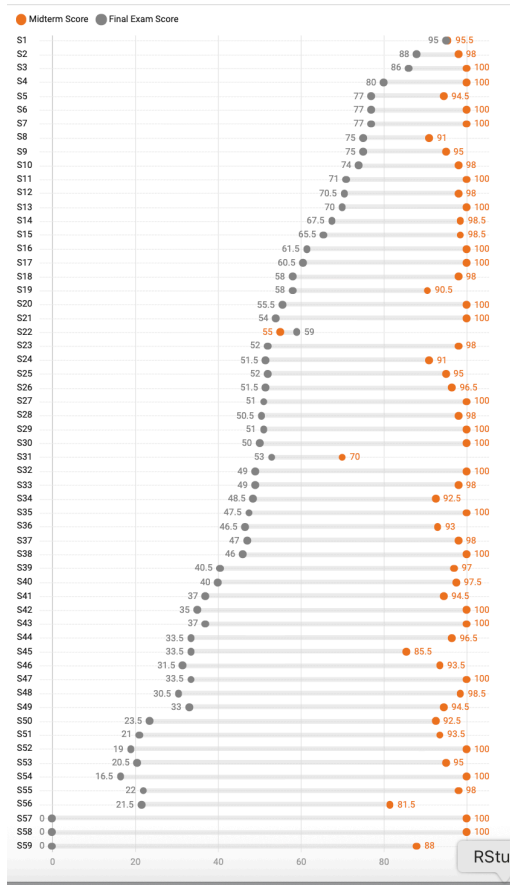
Brown Professor Suspects Majority of His Class Used AI to Cheat

Brown University leaders' response to the alleged cheating incident has been "meek," the professor said.

By [Emma Whitford](#)

"He knew something 'fishy' was going on, and so he and his graders ran the test through ChatGPT. The AI gave answers that mirrored what his students had written, and which were 'kind of correct, but very off and with a very convoluted style,' Serrano said."

Brown Professor Suspects Majority of His Class Used AI to Cheat



Brown Professor Suspects Majority of His Class Used AI to Cheat

Every System Gets the Outcomes It Designs For

There's a maxim from health-care quality

improvement, usually attributed to Paul Batalden:

Every system is perfectly designed to get the results it gets. What I appreciate about that sentence is that it has no villains in it.

Consider one more number from the Brown story.

Serrano's course typically enrolled around 30 students. This spring, with take-home exams promised, 86 signed up. You could read that cynically. Or you could notice that students were choosing a course based on its evaluative architecture, which is exactly what the system has trained them to optimize, in a labor market where grades in certain courses function as tickets to certain careers. Students who treat a grade as a credential to be secured at minimum cost aren't confused about what school is. They've read the design correctly.

And nobody in this story behaved unreasonably. The professor made a humane call after a campus tragedy, checked his evidence, told his students the truth and gave [insidehighered.com](https://www.insidehighered.com) him wrong. The students responded to the incentives in front of

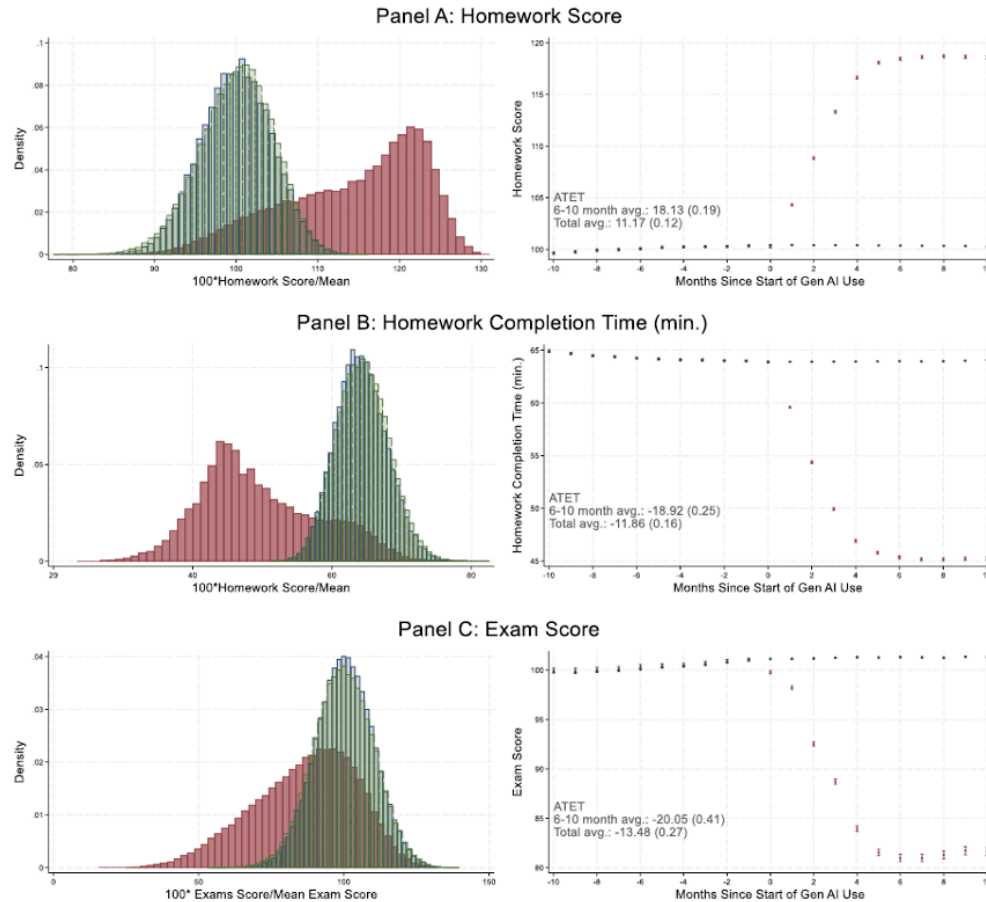
That Chart From Brown Isn't Really About Cheating

The Generative AI Learning Penalty

*"Using 30 months of panel data on 26,811 Chinese students in grades 7-12, we study how generative AI affects homework productivity and learning. The data combine monthly closed-book exams, high-school and college entrance exams, and homework scores and completion time across nine subjects. We exploit staggered AI adoption in a difference-in-differences design. **AI adoption raises homework scores by 18% and reduces completion time by 30%, but lowers monthly exam scores by 20% within six months. High-stakes entrance-exam scores fall by 18 and 24%, with the full penalty emerging only after about two years.** The losses are largest in social science subjects, followed by STEM and languages, and are especially large for junior students, high-achieving students, and boys".*

Stromberg, David and Lei, Victor and Wu, Yanhui, The Generative AI Learning Penalty: Evidence from Chinese Secondary Education (June 02, 2026). Available at SSRN: <https://ssrn.com/abstract=6868618> or <http://dx.doi.org/10.2139/ssrn.6868618>

The Generative AI Learning Penalty



Descriptive Epidemiology

Descriptive Epidemiology

- What is Descriptive Epidemiology?
- How much bias is *acceptable*? (👁️👁️)
- Descriptive vs. Causal (🤔)

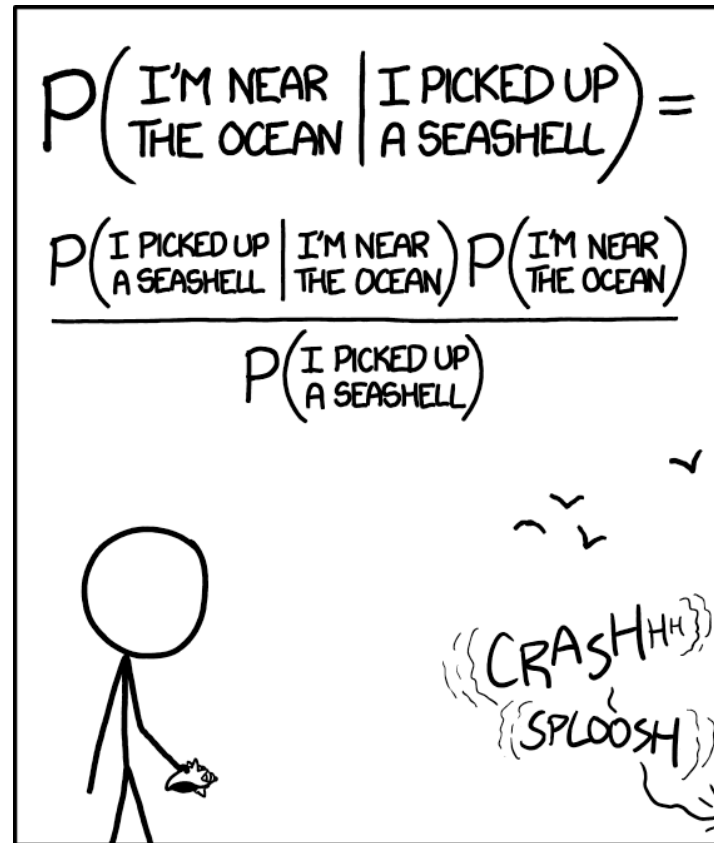
Matthew P Fox, Eleanor J Murray, Catherine R Lesko, Shawnita Sealy-Jefferson, On the Need to Revitalize Descriptive Epidemiology, American Journal of Epidemiology, Volume 191, Issue 7, July 2022, Pages 1174–1179, <https://doi.org/10.1093/aje/kwac056>

**Not every research needs to be causal,
but everything should be robust (!)**

Framework for descriptive Epidemiology

1. Research Question and Background
2. Population and Data Source
3. Outcome Ascertainment/Covariates
4. Analysis Plan: Targeted Measure of Occurrence
5. Results (Presentation of)
6. **Bias, Limitations** and **Interpretations**

A little bit of Bayes



STATISTICALLY SPEAKING, IF YOU PICK UP A SEASHELL AND DON'T HOLD IT TO YOUR EAR, YOU CAN PROBABLY HEAR THE OCEAN.

QUESTIONS?

COMMENTS?

RECOMMENDATIONS?

Baseline Assessment Quizz

EPIB 704 - Baseline (FALL 2026)
(Practice Mode)



Other resources

Reproducible research

- Can you reproduce a project that you completed 3 weeks ago?
- How about 3 months ago?
- 3 years ago?

What if your data changed, even minimally, could you easily redo your analyses?

- How long would it take you to update your manuscript?

Getting setup and starting a project in R Studio

1. Put all the files you received into a folder and give it a name (e.g., `704_course`)
2. Choose `File > New Project`
3. Associate the project with the course folder (e.g., `704_course`)
4. From now on, you can open your project by clicking directly on the `.Rproj` file in that folder (or using the drop-down menu on the top right)
5. Now create an R Notebook file (`File > New File > R Notebook`)

A note on computer code and syntax

- There are two main "dialects" of R: base R and tidyverse
- I generally use tidyverse syntax
- There are three main differences:
 - The pipe `%>%`
 - `dplyr` verbs (e.g., `select()`, `filter()`, `mutate()`)
 - graphing with `ggplot2` (but this antedated the Tidyverse)
- However a case can be made that base R is kinder, gentler and more efficient
- This course is not about learning computer syntax so plenty of sample code will be provided
- It is about understanding what code to use and understanding its output

Tidyverse examples

- Instead of `tidy(lm(y ~ x))`, type `lm(y ~ x) %>% tidy()`
- For data management, you can easily filter and select columns:

```
data_subset <- data %>%  
  filter(female == 0) %>%           # keep female respondents  
  select(income, education) %>%    # keep two variables  
  mutate(educ_2 = education^2) %>% # create edu^2  
  drop_na()                        # listwise delete missings
```

A few useful keyboard shortcuts

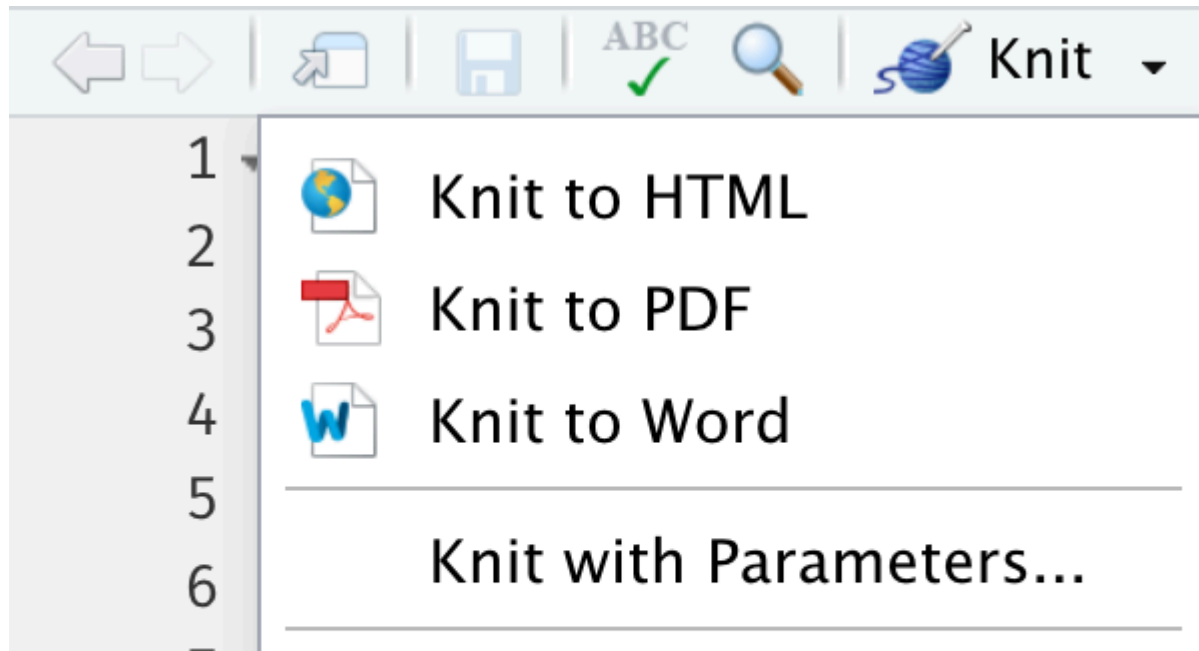
Object/Action	Windows	MacOS
<-	Alt + -	Option + -
%>%	Ctrl + Shift + M	Cmd + Shift + M
New code block	Ctrl + Alt + I	Cmd + Option + I
Run highlight	Ctrl + Enter	Cmd + Enter
Clear console	Ctrl + L	Ctrl + L

R can help reproducibility

Can integrate code and text.

Uses RMarkdown which is simple intuitive language which can be out put to different formats

- HTML
- pdf
- LaTeX
- Word



What is R Markdown?

1. "An authoring framework for data science." (✓)
2. A document format (.Rmd). (✓)
3. An R package named rmarkdown. (✓)
4. "A file format for making dynamic documents with R." (✓)
5. "A tool for integrating text, code, and results." (✓)
6. "A computational document." (✓)
7. Wizardry. (🧙)

R Markdown

R Markdown depends on `knitr` and `Pandoc`

`knitr` executes the computer code embedded in Markdown, and converts R Markdown to Markdown

`Pandoc` renders Markdown to the output format you want (such as PDF, HTML, Word, etc)

- [R Markdown: The Definitive Guide](#)
- `RStudio` has a Markdown quick reference drop down help menu

Basic R-Markdown anatomy

1. The **metadata** (YAML)

2. The **text**

3. The **code**

4. The **output**

Gentle tutorial

R Markdown for writing reproducible scientific papers

R Chunk

```
```{r echo=TRUE, results='hide'}  
glimpse(mockdata)
```
```

default options

```
## List of 54  
## $ eval      : logi TRUE  
## $ echo      : logi FALSE  
## $ results   : chr "markup"  
## $ tidy      : logi FALSE  
## $ tidy.opts : NULL  
## $ collapse  : logi FALSE  
## $ prompt    : logi FALSE  
## $ comment   : chr "##"  
## $ highlight : logi TRUE  
## $ size      : chr "normalsize"  
## $ background : chr "#F7F7F7"  
## $ strip.white : 'AsIs' logi TRUE  
## $ cache      : logi FALSE  
## $ cache.path : chr "L1_EPIB704_26_cache/html/  
## $ cache.vars : NULL  
## $ cache.lazy : logi TRUE  
## $ dependson  : NULL  
## $ autodep    : logi FALSE  
## $ cache.rebuild : logi FALSE  
## $ fig.keep   : chr "high"  
## $ fig.show   : chr "asis"  
## $ fig.align  : chr "center"  
## $ fig.path   : chr "L1_EPIB704_26_files/figure/
```

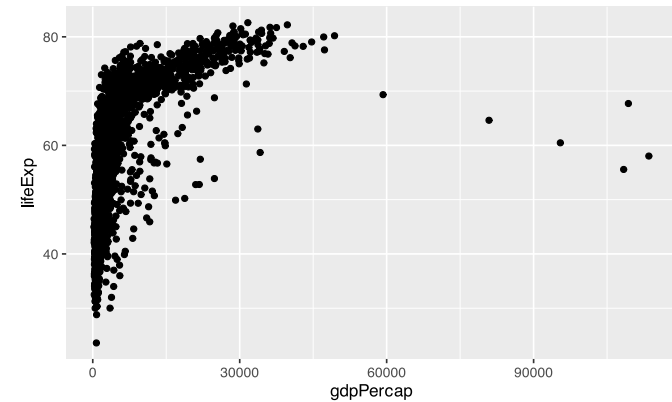
Visualizations

Great way to understand and share insights into your data

R is great for visualizations - either base R or tidyverse with **ggplot2** package

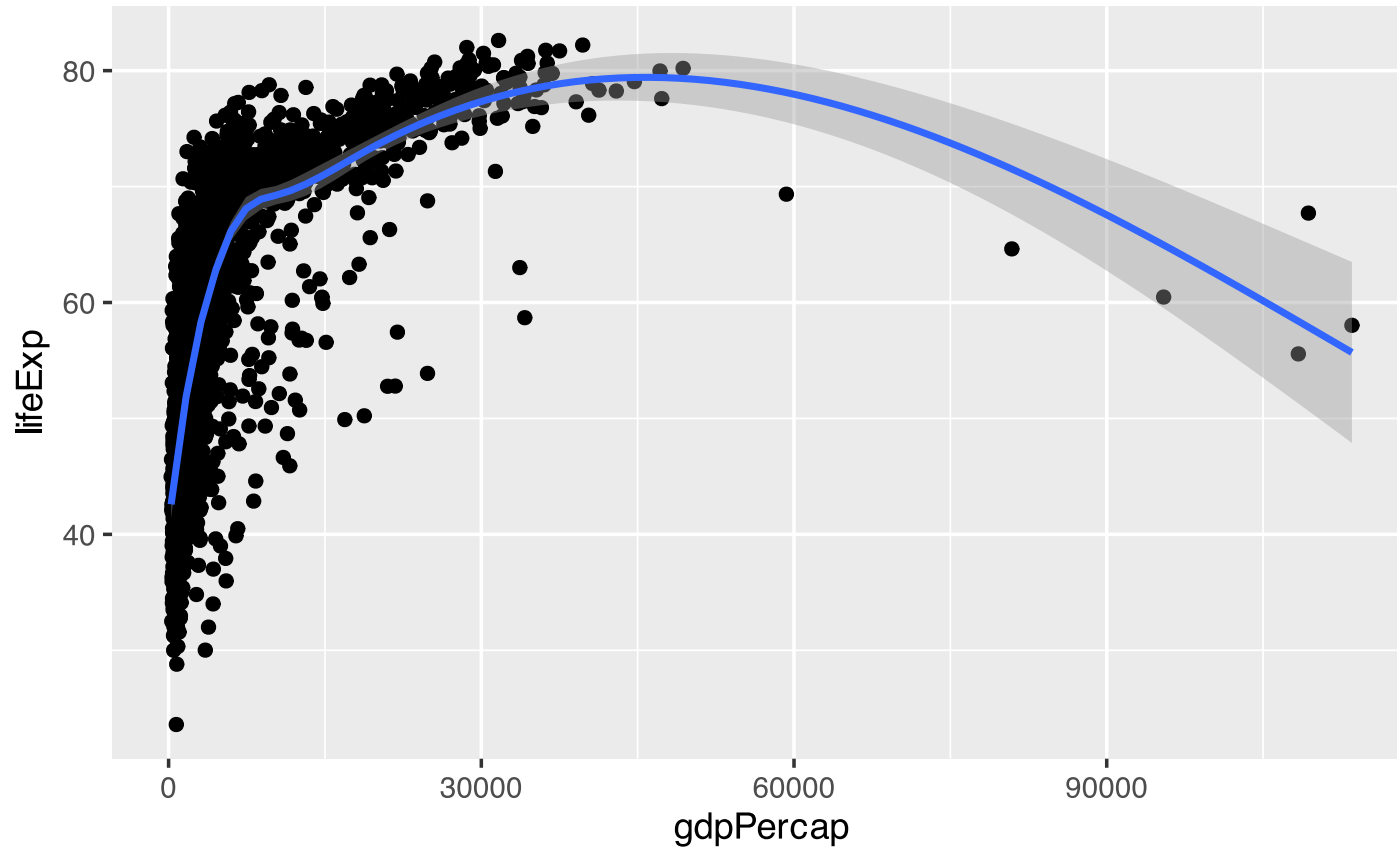
```
library(gapminder)
p <- ggplot(data = gapminder,
            mapping = aes(x = gdpPercap, y =
                          lifeExp))
p <- p + geom_point()
```

p



Build plots with layers

```
p + geom_smooth()
```



Increasing complexity

```
p <- ggplot(data = gapminder,  
           mapping = aes(x = gdpPercap,  
                         y = lifeExp,  
                         color = continent  
                         fill = continent))  
  
geom_point() +  
geom_smooth(method = "loess") +  
scale_x_log10()
```

